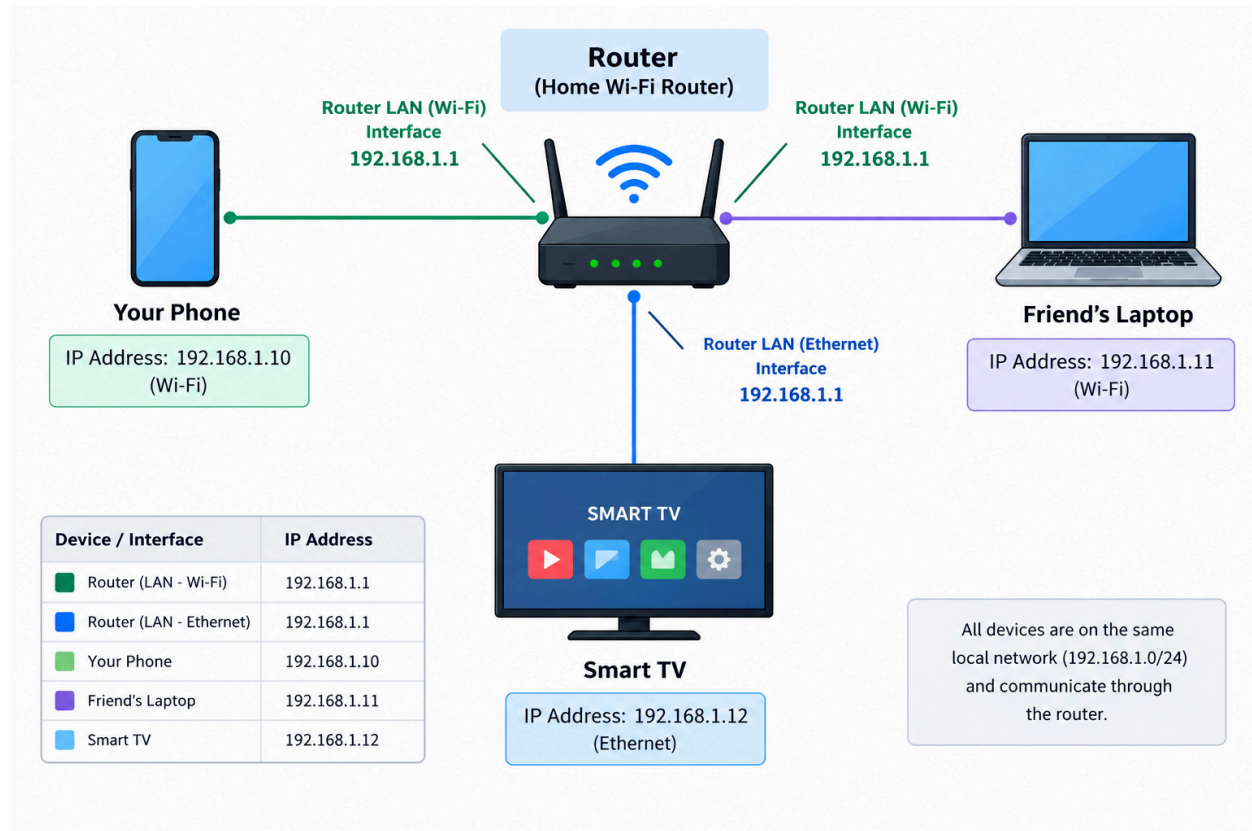


IP ROUTING & ROUTING PROTOCOLS

Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANS:



Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

ANS:

Routing Simulation (Short Version)

- **Source (Your Laptop):** 192.168.1.50 (Gateway: 192.168.1.1)
- **Destination (Friend):** 192.168.2.100

Step-by-Step Path

1. **Laptop Check:** Your laptop compares `192.168.2.100` with its local subnet mask (`255.255.255.0`). Seeing a different network ID (`192.168.2.0` vs `192.168.1.0`), it routes the packet to its **Default Gateway** (`192.168.1.1`).
2. **Router Inspection:** The router receives the packet on Interface A (`192.168.1.1`) and checks its routing table:
 - o **Route Matched:** `192.168.2.0/24` via Direct Connection on Interface B (`192.168.2.1`).
3. **Final Delivery:** The router wraps the packet into a new Ethernet frame and delivers it directly to your friend's MAC address on the destination subnet.

Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

ANS:

Destination	Subnet Mask	Next Hop	Interface	Branch
<code>10.10.0.0</code>	<code>255.255.255.0</code>	Direct	<code>Eth0</code>	HQ Local
<code>10.10.1.0</code>	<code>255.255.255.0</code>	<code>192.168.100.2</code>	<code>WAN1</code>	Mumbai
<code>10.10.2.0</code>	<code>255.255.255.0</code>	<code>192.168.200.2</code>	<code>WAN2</code>	Bangalore
<code>10.10.3.0</code>	<code>255.255.255.0</code>	<code>192.168.220.2</code>	<code>WAN3</code>	Delhi
<code>0.0.0.0</code>	<code>0.0.0.0</code>	<code>203.0.113.1</code>	<code>WAN-ISP</code>	Internet (Default)

Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

ANS:

Here is a comparison of static and dynamic routing, illustrated through the lens of a multi-city food delivery network like Zomato.

1. Static Routing

Manual routes configured by an administrator. Best for small, simple, or fixed networks.

- **Pros:**

-

1. **Zero Overhead:** Consumes no router CPU or bandwidth for route discovery.
- 2.
3. **Predictable & Secure:** Traffic always follows a strictly defined path; no risk of dynamic routing protocol spoofing.
- 4.
5. **Easy for Small Setups:** Simple to implement when there is only one exit point (e.g., a single default gateway to an ISP).
- 6.

- **Cons:**

-

1. **Does Not Scale:** Adding new branch offices or subnets requires manual updates across every router.
- 2.
3. **No Automatic Failover:** If a primary link goes down, traffic drops completely until an admin manually reconfigures the route.
- 4.
5. **High Maintenance:** Prone to human error when managing large, growing networks.
- 6.

- **Real-World Example:** A single small-town Zomato partner restaurant with one broadband router connecting its local order-printing tablet directly to the cloud.

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2. Dynamic Routing (OSPF, BGP, RIP)

Routers automatically discover neighbors, exchange topology updates, and calculate the best paths using algorithms.

- **Pros:**

-

1. **Automatic Failover:** If a fiber line between the Mumbai data center and Bangalore hub is cut, protocols like OSPF instantly route traffic through an alternative path with zero human intervention.
 - 2.
 3. **High Scalability:** New restaurant branches or cloud servers automatically announce themselves to the network.
 - 4.
 5. **Optimized Paths:** Constantly recalculates the fastest route based on real-time link costs and congestion.
 - 6.
- **Cons:**
 -
1. **Resource Intensive:** Consumes router CPU, memory, and bandwidth to continuously send hello packets and routing updates.
 - 2.
 3. **Configuration Complexity:** Requires specialized knowledge to set up, tune timers, and troubleshoot protocol states.
 - 4.
 5. **Security Vulnerabilities:** Misconfigurations or malicious injections can cause routing loops or blackhole traffic (e.g., BGP hijacking).
 - 6.
- **Real-World Example:** Zomato's multi-region cloud architecture (AWS/GCP zones) across different cities, where dynamic routing ensures food orders, delivery partner GPS pings, and restaurant updates find the fastest available path instantly.

Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

ANS:

Short OSPF Configuration & Breakdown

Sample Config (Central Router):

```
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

What Each Line Does:

- **router ospf 1:** Enables the OSPF routing process.

- `router-id 1.1.1.1`: Assigns a unique identifier to the router.
- `network 192.168.1.0 0.0.0.255 area 0`: Tells OSPF to advertise this local subnet using a wildcard mask (`0.0.0.255`) into the core **Backbone Area 0**.
- `network 10.0.0.0 0.0.0.3 area 0`: Advertises the point-to-point link connecting to neighboring routers.